

Deterministic geometric Savitzky-Golay dynamics estimator

Shared parameter update: reference -> estimated

mass [kg]	2.351597591	->	2.351597591	delta=+3.07723e-11	ratio=1
CoG x [m]	-0.002024708562	->	-0.002024708594	delta=-3.2194e-11	
ratio=-					
CoG y [m]	-3.052657894e-05	->	-3.052663663e-05	delta=-5.76891e-11	
ratio=-					
CoG z [m]	0.009509749599	->	0.009509749675	delta=+7.59653e-11	
ratio=-					
rotor effectiveness 1	1	->	0.9999999999	delta=-5.3436e-11	ratio=1
rotor effectiveness 2	1	->	0.9999999999	delta=-6.6985e-11	ratio=1
rotor effectiveness 3	1	->	0.9999999999	delta=-6.36398e-11	ratio=1
rotor effectiveness 4	1	->	0.9999999999	delta=-6.4204e-11	ratio=1
command lag [s]	0	->	0.0040000000385	delta=+0.004	ratio=-

Inertia matrix [kg m^2]

reference:

[0.06500006148, -7.27899253e-07, 1.901508003e-05]
[-7.27899253e-07, 0.06495265634, 5.9167305e-08]
[1.901508003e-05, 5.9167305e-08, 0.1289921107]

estimated:

[0.06500006148, -7.278992935e-07, 1.901507951e-05]
[-7.278992935e-07, 0.06495265634, 5.916651423e-08]
[1.901507951e-05, 5.916651423e-08, 0.1289921107]

Principal moments: reference -> estimated

principal moment 1 [kg m^2]	0.06495264516	->	0.06495264517	delta=+2.39879e-12	
ratio=1					
principal moment 2 [kg m^2]	0.06500006701	->	0.06500006701	delta=+8.83488e-13	
ratio=1					
principal moment 3 [kg m^2]	0.1289921163	->	0.1289921163	delta=+7.93005e-13	ratio=1

Optimization

joint dynamics loss: 927.026962454

Gaussian-prior cost: 5.18026029722e-17

selected strict-ZOH lag [s]: 0.0040000000385

Bag failure_1_single

reconstruction position RMSE [m]: 88.0913549914

reconstruction position component RMSE [m]: [16.173556 2.49224 86.558025]

reconstruction orientation RMSE [deg]: 151.46977662

gyro consistency RMSE xyz [rad/s]: [56.148377 58.085965 21.198649]

specific-force consistency RMSE xyz [m/s^2]: [145.255666 121.371031 396.425954]

inferred external force RMS xyz [N]: [17.644999 13.804448 12.992707]

inferred external torque RMS xyz [N m]: [4.57309 6.473208 9.601725]

bag dynamics loss: 927.026962454

Recorded command timestamp intervals (data-derived)

Bag failure_1_single

rotor: median dt=0.00499773026 s (200.091 Hz), min=2.00271606e-05 s, max=0.0138893127 s

gimbal: median dt=0.00499868393 s (200.053 Hz), min=4.529953e-06 s, max=2.48282003 s